Sefat E Rahman

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Education

The University of Utah – *Ph.D. in Computer Science and Engineering* – Salt Lake City, UT — 2023 – Present

University of South Florida – *M.S. in Computer Science and Engineering* – Tampa, FL — 2021 – 2022

Khulna University of Engineering & Technology – *B.S. in Computer Science and Engineering* – Khulna, Bangladesh — 2013 – 2018

Experience

Lawrence Berkeley National Laboratory – Machine Learning Research Intern – Berkeley, CA — Summer 2026

- Studied the **global loss landscapes** of molecular property prediction models on the **ESOL** solubility dataset, using **mode connectivity** to test whether independently trained solutions are joined by low-loss paths.
- Characterized basins, barriers, and local curvature of the weight space with **Mapper**, **merge trees**, and **Hessian eigenvalue** analysis.
- Ran large-scale training and connectivity experiments on the **NERSC** supercomputer.

The University of Utah – Graduate Research Assistant – Salt Lake City, UT — 2023 – Present

- Developed **GASP**, an open-source algorithm for faithful Reeb graph visualization, benchmarked against TTK (**TopoInVis 2025**).
- Developing **PlotPreserve**, a framework for **scatterplot modification** that preserves **data properties** (statistical, structural, distributional) while points are reduced, perturbed, or transitioned.

The University of Utah – Graduate Teaching Assistant – Salt Lake City, UT — Fall 2023, 2025, 2026

- Supported the **Visualization for Data Science** course and delivered a guest lecture on **Text Visualization**.

University of South Florida – Graduate Teaching Assistant – Tampa, FL — 2021 – 2022

- Led **Computer Architecture** and **Computer Design** labs; assisted with Cloud Computing and Programming Concepts.

Eastern University – Lecturer – Dhaka, Bangladesh — 2018 – 2020

- Primary instructor for undergraduate **DBMS** and **Machine Learning**; supervised theses and capstone projects.

Peer-Reviewed Publications • 1st author

- GASP: A Gradient-Aware Shortest Path Algorithm for Boundary-Confined 2-Manifold Reeb Graph Visualization. ***IEEE TopoInVis 2025***.
- GASP: A Gradient-Aware Shortest Path Algorithm for Boundary-Confined Visualization of 3D Reeb Graphs. ***IEEE VIS 2024*** (Poster).
- Email Spam Detection using Bidirectional Long Short Term Memory with Convolutional Neural Network. ***IEEE TENSYMP 2020***.

Manuscripts in Preparation

Visual Analytics of Mode Connectivity in ML Loss Landscapes | *Mapper, Merge Trees, Dash, Plotly* — *In preparation: EuroVis 2027*

- Built an interactive **visual analytics dashboard** to compare loss landscapes across data splits and train/validation/test metrics.
- Linked **Mapper** graphs, **merge trees**, **mode connectivity** paths, and local **Hessian** spectra in coordinated views.

PlotPreserve | *Persistent Homology, Wasserstein & Bottleneck Distances, Optimal Transport* — *In revision: IEEE TVCG (Dec 2026)*

- Designed **PlotPreserve**, a unified framework for **scatterplot modification** that preserves **data properties** (statistical, structural, distributional) while points are added, removed, or moved, applied to adaptive visualization, privacy, and dataset interpolation.

Uncertainty-Aware Visualization of Reeb Graphs | *Reeb Graphs, Edge Bundling* — *In preparation: IEEE VIS 2027*

- Visualizing uncertain Reeb graphs from noisy scalar fields with **edge bundling** to reduce clutter and improve interpretability.

Projects

Assign | *Amazon Bedrock, ECS Fargate, Aurora Serverless, Zod* | **AI in Education Hackathon Honorable Mention** (26 teams) — 2026

- Co-developed an AI assignment platform that diagnoses the gap behind each wrong answer, reteaches it, and verifies mastery.
- Five AI agents (quiz, grader, tutor, chat, insights) with **Zod**-validated outputs and guardrails; teachers get **concept-level evidence**.

Technical Skills

Generative AI: Multi-Agent Systems, LLM Guardrails, RAG, Amazon Bedrock, Claude Code, Codex, Agent Harnesses
Languages & Tools: Python, C/C++, JavaScript, Node.js, SQL, HTML/CSS, Git, uv, AWS (ECS Fargate, Aurora), NERSC HPC
Visualization: ParaView, VTK, MeshLab, Blender, D3.js, Dash, Plotly, Surface (OBJ) & Volumetric (VTK) Meshes
Machine Learning: PyTorch, Scikit-learn, TensorFlow, MATLAB

Leadership

Bangladesh Student Association, The University of Utah (ASUU) – Vice President – Salt Lake City, UT — 2023 – 2024